Project Development Intimation Document

Project: REB AMI (Advanced Metering Infrastructure)

Organization: OTBL (Onnorokom Technology Bangladesh Limited)

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1. Project Overview

1.1 Project Title

REB AMI — Advanced Metering Infrastructure System

1.2 Project Sponsor

Rural Electrification Board (REB), Bangladesh

1.3 Development Partner

Onnorokom Technology Bangladesh Limited (OTBL)

1.4 Document Version

Version	Date	Author	Description
1.0	April 2026	OTBL Dev Team	Initial Release

1.5 Project Summary

The REB AMI project aims to develop a comprehensive Advanced Metering Infrastructure system to modernize electricity metering, billing, and data management for Rural Electrification Board. The system will enable real-time meter data collection, automated billing, outage detection, and analytics-driven decision making.

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2. Project Objectives

2.1 Primary Objectives

- Implement automated meter reading (AMR) capabilities across REB service areas
- Develop a centralized data management platform for meter data collection and processing

- Create real-time monitoring dashboards for grid operators and administrators
- Enable mobile accessibility for field personnel and consumers
- Integrate with existing Oracle-based billing and ERP systems

2.2 Secondary Objectives

- Reduce manual meter reading costs by 80%
- Minimize billing errors and revenue leakage
- Enable predictive maintenance through data analytics
- Improve customer satisfaction through transparent billing

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3. Scope of Development

3.1 In-Scope Items

Module	Description	Priority
Meter Data Management	Collection, validation, and storage of meter readings	High
Billing Integration	Automated billing calculation and Oracle sync	High
Web Dashboard	Vue.js-based administrative dashboard	High
Mobile Application	Android/iOS app for field staff and consumers	Medium
Analytics Engine	Reporting and predictive analytics	Medium
API Gateway	RESTful APIs for third-party integration	High

3.2 Out-of-Scope Items

- Hardware procurement for smart meters (handled by REB)
- Network infrastructure deployment (handled by REB IT)
- Legacy system decommissioning
- End-user training (separate contract)

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4. Technical Architecture Summary

4.1 Technology Stack

Layer	Technology	Justification
Frontend (Web)	Vue.js 3.x	Reactive UI, component-based architecture
Frontend (Mobile)	React Native	Cross-platform iOS/Android support
Backend	Node.js (Express.js)	High performance, event-driven I/O
Database	PostgreSQL 15	ACID compliance, JSON support, scalability
Caching	Redis	In-memory caching for performance
Load Balancer	NGINX	Reverse proxy, SSL termination
External Integration	Oracle DB	Legacy billing system integration

4.2 Infrastructure Requirements

Component	Specification	Quantity
Application Server VMs	8 vCPU, 32GB RAM, 500GB SSD	2
Database Server	16 vCPU, 64GB RAM, 2TB SSD	1
Web Server VM	4 vCPU, 16GB RAM, 200GB SSD	1
Load Balancer	NGINX Plus	1

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5. Development Timeline

5.1 Project Phases

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Phase 1: Foundation (Weeks 1-6)
├── Requirements finalization
├── Architecture design approval
├── Development environment setup
└── Database schema design

Phase 2: Core Development (Weeks 7-18)
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├── Backend API development
├── Database implementation
├── Vue.js dashboard development
└── Mobile app development

Phase 3: Integration (Weeks 19-24)
├── Oracle system integration
├── NGINX configuration
├── End-to-end testing
└── Performance optimization

Phase 4: Deployment (Weeks 25-28)
├── UAT environment deployment
├── User acceptance testing
├── Production deployment
└── Go-live support
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5.2 Key Milestones

Milestone	Target Date	Deliverable
M1: Design Approval	Week 4	Approved DRD & Architecture
M2: API Completion	Week 14	All backend APIs functional
M3: Dashboard Beta	Week 18	Vue.js dashboard ready for UAT
M4: Integration Complete	Week 24	Oracle sync operational
M5: Go-Live	Week 28	Production deployment

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6. Team Structure

6.1 Project Team Composition

Role	Count	Responsibility
Project Manager	1	Overall project coordination
Technical Lead	1	Architecture & technical decisions

Role	Count	Responsibility
Backend Developers	3	Node.js API development
Frontend Developers	2	Vue.js & React Native development
Database Engineer	1	PostgreSQL & Oracle integration
QA Engineers	2	Testing & quality assurance
DevOps Engineer	1	CI/CD, deployment, monitoring

6.2 RACI Matrix

Activity	PM	Tech Lead	Dev	QA	Client
Requirements	A	C	I	I	R
Architecture	A	R	C	I	C
Development	I	A	R	C	I
Testing	I	C	C	R	A
Deployment	A	R	C	C	I

R=Responsible, A=Accountable, C=Consulted, I=Informed

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7. Risk Assessment

7.1 Identified Risks

Risk ID	Description	Probability	Impact	Mitigation
R1	Oracle integration complexity	High	High	Early POC, dedicated resource
R2	Data migration challenges	Medium	High	Phased migration approach
R3	Network latency in rural areas	Medium	Medium	Offline-capable mobile app
R4	Resource availability	Low	Medium	Cross-training team members
R5	Scope creep	Medium	High	Strict change control process

8. Communication Plan

8.1 Stakeholder Communication

Stakeholder	Frequency	Method	Content
REB Management	Bi-weekly	Presentation	Progress, risks, decisions
Technical Team	Daily	Stand-up	Task status, blockers
QA Team	Weekly	Meeting	Test results, defects
End Users	Monthly	Newsletter	Feature updates

9. Quality Assurance

9.1 Quality Standards

- Code review mandatory for all commits
- Minimum 80% unit test coverage
- Automated integration testing
- Performance benchmarks: API response < 200ms
- Security audit before production deployment

9.2 Testing Strategy

- Unit Testing: Jest (Backend), Vue Test Utils (Frontend)
- Integration Testing: Postman/Newman
- Load Testing: Apache JMeter
- Security Testing: OWASP ZAP

10. Approval & Sign-off

10.1 Document Approval

Role	Name	Signature	Date
Project Sponsor (REB)	_____	_____	_____
Project Manager (OTBL)	_____	_____	_____
Technical Lead (OTBL)	_____	_____	_____

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